7 Feedback Amplifier

INTRODUCTION

A feedback amplifier is an amplifier in which the output signal is sampled, and the output signal is given as a feedback to the input in order to produce an error signal which collectively forms an amplifier.

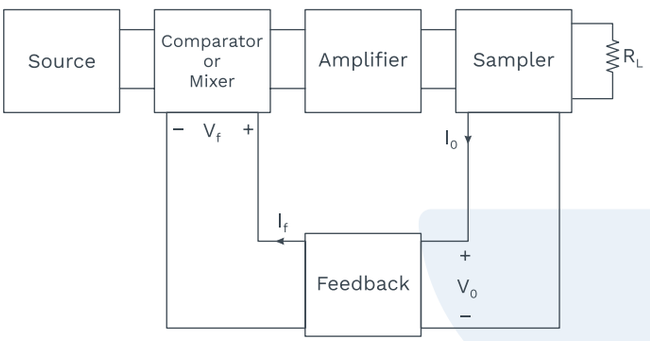


Fig. 7.1 Basic Feedback Amplifier

Feedback can be positive or negative but in the case of an amplifier, negative feedback is used.

Example 1:
Why is negative feedback used?

Sol:

- To stabilize the voltage gain.
- To increase the bandwidth.
- To increase the input impedance.
- Linear operation.
- To reduce output impedance.
- To reduce noise.

Sampling Circuit:

- This circuit is used to sample the output signal and connect it to the feedback network.
- There are two types of sampling circuits:
  - **a)** Shunt sampling or voltage sampling
  - **b)** Series sampling or current sampling

Grey Matter Alert!

In positive feedback, input signal gets added to the feedback signal, whereas in a negative feedback, the feedback signal gets subtracted from the input signal.

Rack Your Brain

What is the need for a feedback from the control system point of view?

a) Shunt sampling or voltage sampling:

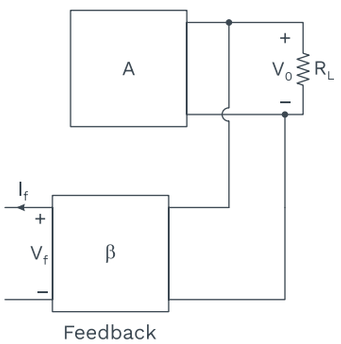


Fig. 7.2 Shunt Sampling

The input terminals of the feedback element are connected in parallel to output, so it is called voltage sampling.
The feedback element has its output as a voltage or current.

So, $\frac{V_f}{V_0} = \beta$ (Unitless)

$\frac{I_f}{V_0} = \beta \, \mho$

b) Current sampling or series sampling:

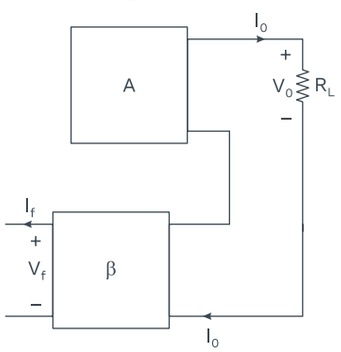


Fig. 7.3 Series Sampling

- The output current of an amplifier acts as input current to the feedback element and hence feedback is connected in series with the output.

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What is the difference between a positive and a negative feedback?

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The physical signals are not current or voltage but converted into current or voltage signal so that signal transmission would be easy.

Previous Years' Question

Negative feedback in an amplifier-
a) reduces gain
b) increases frequency and phase distortions
c) reduces bandwidth
d) increases noise

- If the feedback depends on load current then, it is called current sampling.

$$\frac{I_f}{I_0} = \beta \quad \text{(Unitless)}$$

$$\frac{V_f}{I_0} = \beta \quad (\Omega)$$

Mixer network:

There are two types of mixer networks-

a) Series mixing

b) Shunt mixing

a) Series Mixing

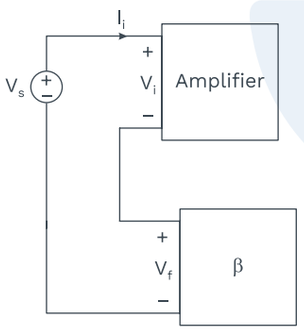


Fig. 7.4 Series Mixing

- It is also called as voltage mixing.

  By KVL, $V_s - V_i - V_f = 0$

$$V_i = V_s - V_f$$

- The source is not directly fed to the amplifier but the feedback voltage is subtracted from it.

b) Shunt mixing:

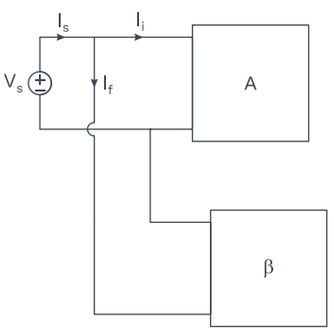


Fig. 7.5 Shunt Mixing

Rack Your Brain

State the difference between 'open-loop gain' and 'closed-loop gain'.

Grey Matter Alert!

The negative feedback decreases the gain of the amplifier. Therefore as the bandwidth increases, the product of gain and bandwidth is always constant.

- It is also known as current mixing.
- The output terminals of feedback are connected in parallel to source so that the current of source and feedback gets mixed.

By KCL, $I_i = I_s - I_f$

- The source current is not fed to the amplifier directly, the feedback current is subtracted from the source current and the resulting current is fed to the amplifier.

Rack Your Brain

Which system has a higher sensitivity, positive feedback or negative feedback?

Closed loop gain of an amplifier:

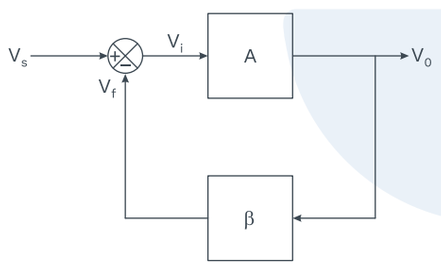


Fig. 7.6 Closed loop system

- Due to feedback element,

$$V_f = \beta V_0$$

The input voltage to the amplifier is given by:

$$V_i = V_s - V_f$$

$$V_i = V_s - \beta V_0$$

The output of the amplifier is:

$$V_0 = AV_i$$

$$V_0 = A(V_s - \beta V_0)$$

$$V_0 = AV_s - A\beta V_0$$

$$\frac{V_0}{V_s} = \frac{A}{1 + A\beta} = A_f$$

Negative feedback is also called degenerative feedback and is given by-

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The feedback element is used as a sensor which calculates the output signal and sends it back to the error detector.

$$N = 20\log_{10}\left|\frac{A_f}{A}\right| = 20\log_{10}\left|\frac{1}{1+A\beta}\right|$$

where N is the degenerative factor.

1) For positive feedback:

In positive feedback, the feedback signal is added to the input signal.

The input to the amplifier is:

$$V_i = V_s + V_f$$

$$V_i = V_s + \beta V_0$$

The output of the amplifier is given by:

$$V_0 = AV_i$$

$$V_0 = A(V_s + \beta V_0)$$

$$V_0(1 - A\beta) = AV_s$$

$$\frac{V_0}{V_s} = \frac{A}{1-A\beta} = A_f$$

where, $A_f > A$

Positive feedback is also called as regenerative feedback.

2) For negative feedback:

$$\text{Gain} = \frac{A}{1+A\beta} = A_f$$

$A_f < A$ Degenerative feedback

Gain sensitivity:

1) The sensitivity indicates the change in the closed-loop gain of the amplifier with respect to the change in open-loop gain.

2) The sensitivity of an amplifier should be as low as possible because if the open-loop gain changes due to the external environment, the closed-loop gain should remain the same.

We know that,

$$A_f = \frac{A}{1+A\beta}$$



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What is the main disadvantage of negative feedback?

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The sensitivity of the feedback element should be as low as possible due to the external environment.

Differentiate feedback gain (A_f) with respect to A, we get-

$$\frac{dA_f}{dA} = \frac{(1+A\beta) - A\beta}{(1+A\beta)^2}$$

$$\frac{dA_f}{dA} = \frac{1}{(1+A\beta)^2}$$

$$\text{Sensitivity} = \frac{dA_f/A_f}{dA/A}$$

$$S = \frac{dA_f}{dA} \times \frac{A}{A_f}$$

$$S = \frac{1}{(1+A\beta)^2} \times \frac{A}{A} \times (1+A\beta)$$

$$S = \frac{1}{(1+A\beta)}$$

$$S = \frac{1}{1+A\beta} = \text{Ratio of percentage change in } A_f \text{ to percentage change in A}$$

$$\%\ \text{change in}\ A_f = \frac{1}{(1+A\beta)} \times \%\ \text{change in A}$$

- De-sensitivity $D = \frac{1}{S} = 1+A\beta$
- The sensitivity should be as low as possible and de-sensitivity should be as large as possible.

Effect of negative feedback on an amplifier:

a) Frequency distortion:

- If the amplifier contains reactive elements, then gain is the function of frequency.
- Due to negative feedback, the effect of change of open-loop gain is minimized so the frequency distortion is reduced.

b) Non-linear distortion:

The non-linear distortion is reduced by a factor of 1+Aβ.

$$D_{nf} = \frac{D_n}{1+A\beta}$$

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The feedback element used in the process industries are thermocouple, thermistor, strain gauge etc.

Rack Your Brain

What is the advantage of increasing bandwidth of any system?

where, D_{nf} = Non-linear distortion with negative feedback

D_n = Non-linear distortion

Noise:

- Noise is a signal which produces an unwanted effect and degrades the quality of the signal.

Internal noise: Noise generated due to elements within the system is termed as internal noise.

External noise: Noise generated due to elements outside the system is termed as the external noise.

$$V_0 = AV_i + V_N + V_D$$

where, V_N = Noise

V_D = Distortion

- Noise is an unwanted external signal added to the original signal, whereas distortion is the change in the shape of the original signal.

$$V_0 = A(V_s - V_f) + V_N + V_D$$

$$V_0 = A(V_s - \beta V_0) + V_N + V_D$$

$$V_0 = \frac{A}{1+A\beta}V_s + \frac{V_N}{1+A\beta} + \frac{V_D}{1+A\beta}$$

Bandwidth:

At low frequency, the open loop frequency response is:

$$OL(A)_{LF} = \frac{A}{1 - j(f_L/f)} \quad \{\text{High pass filter}\}$$

Frequency response with feedback is-

$$(A_f)_{LF} = \frac{A_{LF}}{1 + A_{LF}\beta}$$

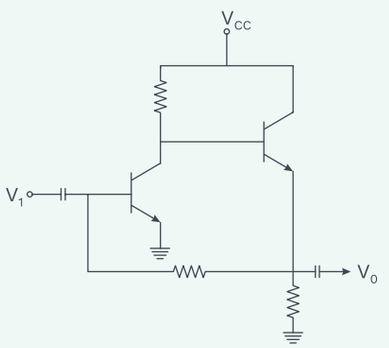
$$(A_f)_{LF} = \frac{\frac{A}{1 - j(f_L/f)}}{1 + \frac{A\beta}{1 - j(f_L/f)}}$$

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The input of the feedback element decides which type of sampling is done.

Previous Years' Question

The feedback amplifier shown in figure has-



a) current-series feedback with large input impedance and large output impedance

b) voltage-series feedback with large input impedance and low output impedance

c) voltage-shunt feedback with low input impedance and low output impedance

d) current-shunt feedback with low input impedance and large output impedance

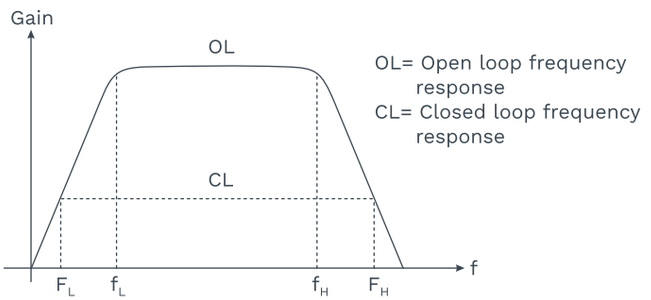


Fig. 7.7 Bandwidth of Open loop and Closed loop

$$(A_f)_{LF} = \frac{A}{1 - j(f_L/f) + A\beta} = \left(\frac{A}{1 + A\beta}\right) \times \frac{1}{1 - j\left(\frac{f_L}{f(1 + A\beta)}\right)}$$

$$(A_f)_{LF} = \frac{A_f}{1 - j\frac{F_L}{f}}$$

where, $A_f = \frac{A}{1 + A\beta}$, $F_L = \frac{f_L}{1 + A\beta}$

$F_L < f_L$ due to negative feedback, lower cut-off frequency of system reduces.

At high frequency, open loop frequency response is:

$$(A)_{HF} = \frac{A}{1 + j(f/f_H)} \quad \{\text{Low pass filter}\}$$

Due to closed loop,

$$(A_f)_{HF} = \frac{(A)_{HF}}{1 + (A)_{HF}\beta}$$

$$(A_f)_{HF} = \frac{A}{1 + j(f/f_H) + A\beta}$$

$$(A_f)_{HF} = \frac{\frac{A}{1 + A\beta}}{1 + j\frac{f}{f_H(1 + A\beta)}}$$

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Which parameter is stabilized in a current shunt feedback?

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In control system applications, negative feedback is used.

$$(A_f)_{HF} = \frac{A_f}{1 + j(f/F_H)}$$

where, $A_f = \frac{A}{1 + A\beta}$

$F_H = f_H(1 + A\beta)$

Due to negative feedback, the upper cut-off frequency of system increases.

$(BW)_{OL} = f_H - f_L \approx f_H (f_H >> f_L)$

$(BW)_{CL} = F_H - F_L \approx F_H \quad (F_H >> F_L)$

$(BW)_{CL} = f_H(1 + A\beta) = (BW)_{OL}(1 + A\beta)$

$\therefore$ Bandwidth increases by factor of $(1 + A\beta)$

How to identify natives of feedback?

- At the input side, check if the input voltage has feedback term or input current has feedback.
  Current mixing: $V_i = V_s$; $I_i = I_s - I_f$ (By KCL)
  Voltage mixing: $I_i = I_s$; $V_i = V_s - V_f$ (By KVL)
- In case of voltage mixing, make $I_i = 0$
  if $V_f \propto V_0$: Voltage sampling
  $V_f \propto I_0$: Current sampling

Proportionality constant must not contain R_L.

- In case of current mixing, make $V_i = 0$
  if $I_f \propto V_0$: Voltage sampling
  $I_f \propto I_0$: Current sampling

Proportionality constant must not contain R_L.

Example 2:

What is the effect of negative feedback on the amplifier?

a) There is a decrease in the gain of the amplifier.
b) There is an increase in frequency.
c) There is a decrease in bandwidth.
d) There is an increase in phase distortion.

Sol:

We know that, $A_f = \frac{A}{1 + A\beta}$

Negative feedback reduces gain.
Hence, the correct option is **a)**.

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In current shunt feedback, the stabilized parameter is current gain.

FEEDBACK TOPOLOGY

There are four types of feedback topology:

1) Series–shunt feedback or voltage– series feedback
(Mixing, Sampling — Sampling, Mixing)

2) Shunt-shunt feedback or voltage-shunt feedback

3) Series-series feedback or current-series feedback

4) Shunt-series feedback or current-shunt feedback

1) Series-shunt feedback:

This is also known as voltage-series feedback.

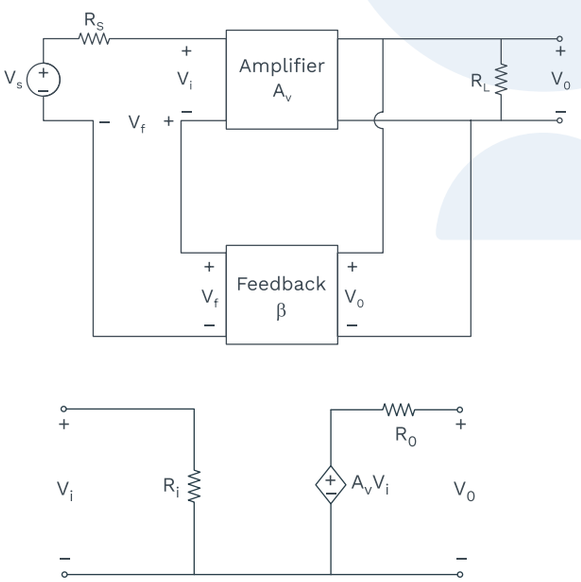


Fig. 7.8 (a) Series-Shunt with Equivalent Circuit

R_i: Input resistance of amplifier
R_0: Output resistance of amplifier
A_v: Voltage gain of amplifier

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In the collector to base bias circuit, the resistance (R_B) acts as the feedback element.

Previous Years' Question

An amplifier without feedback has a voltage gain of 50, input resistance of 1 kΩ and output resistance of 2.5 kΩ. The input resistance of the current-shunt negative feedback amplifier using the above amplifier with a feedback factor of 0.2 is:

a) 1/11 kΩ **b)** 1/5 kΩ
c) 5 kΩ **d)** 11 kΩ

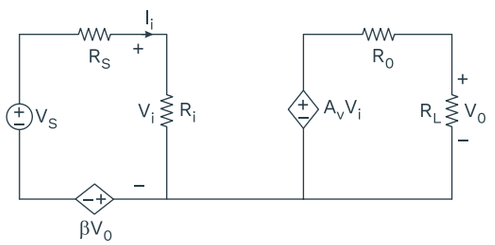


Fig. 7.8 (b) Equivalent circuit with source and load

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In series-series feedback, the feedback input is current.

Applying KVL in the input loop, we get-

$$V_S - I_i R_S - V_i - \beta V_0 = 0 \qquad ...(i)$$

By voltage division rule:

$$V_0 = A_v V_i \frac{R_L}{R_0 + R_L} = A_{VL} V_i \qquad ...(ii)$$

where, $A_{VL} = \dfrac{A_v \times R_L}{R_0 + R_L}$

A_{VL} = voltage gain considering R_L

Putting the value of V_0 in the equation (i), we get-

$$V_S - I_i R_s - I_i R_i - A_{VL}\beta V_i = 0$$

$$V_S - I_i R_s - I_i R_i - A_{VL}\beta I_i R_i = 0$$

$$\therefore \quad V_s = I_i R_s + I_i R_i (1 + A_{VL}\beta)$$

Input resistance:

Input resistance with feedback is-

$$R_{if} = R_i(1 + A_{VL}\beta)$$

where, $A_{VL} = \dfrac{A_v \cdot R_L}{R_0 + R_L}$

Output resistance:

Without feedback: $V_0 = A_v V_i - I_0 R_0$

With feedback: $V_i = \dfrac{(V_s - V_f)R_i}{R_i + R_s}$

$$V_0 = \frac{A_v R_i}{R_i + R_s}(V_s - \beta V_0) - I_0 R_0 \qquad \{V_f = \beta V_0\}$$



Rack Your Brain

Which type of feedback is used in the oscillator circuit?

$$V_0 = A_{vs}(V_s - \beta V_0) - I_0 R_0$$

$$V_0(1 + A_{vs}\beta) = A_{vs} V_s - I_0 R_0$$

$$V_0 = \frac{A_{vs}}{1 + A_{vs}\beta} V_s - \frac{I_0 R_0}{1 + A_{vs}\beta}$$

A_{vs}: Voltage gain considering source resistance

where, $A_{VS} = \dfrac{A_V R_i}{R_i + R_S}$

Output resistance with feedback is-

$$R_{0f} = \frac{R_0}{1 + A_{vs}\beta}$$

For example: Common collector amplifier

2) Shunt-shunt feedback:

This is also known as voltage-shunt feedback.

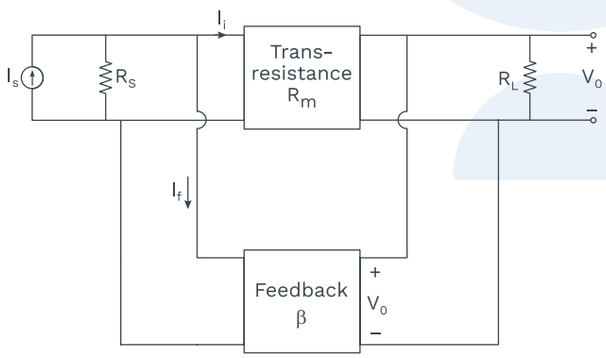


Fig. 7.9 (a) Shunt-Shunt Feedback Topology

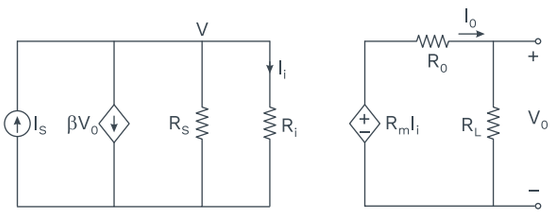


Fig. 7.9 (b) Shunt-Shunt Feedback Topology

$$V_0 = R_m I_i \left(\frac{R_L}{R_0 + R_L} \right) = R_{mL} I_i$$

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The current shunt feedback circuit is also known as shunt-series feedback.

where, $R_{mL} = \frac{R_m \cdot R_L}{R_0 + R_L}$

R_{mL}: Trans-resistance considering load resistance
Applying KCL at node V, we get-

$$I_s - \beta V_0 = \frac{V}{R_s} + \frac{V}{R_i}$$

$$I_s - \beta R_{mL} I_i = \frac{V}{R_s} + \frac{V}{R_i}$$

$$I_s - \beta R_{mL} \frac{V}{R_i} = \frac{V}{R_s} + \frac{V}{R_i}$$

$$I_s = \frac{V}{R_s} + \frac{V}{R_i}(1 + R_{mL}\beta)$$

Input resistance:

$$R_{if} = \frac{R_i}{1 + R_{mL}\beta}$$

Output resistance:

Applying KVL in the output loop, we get-

$$R_m I_i - I_0 R_0 = V_0$$

$$R_m \left[\frac{(I_S - \beta V_0) \times R_S}{R_i + R_s}\right] - I_0 R_0 = V_0$$

$$R_{ms}(I_s - \beta V_0) - I_0 R_0 = V_0$$

where, $R_{ms} = \frac{R_m R_S}{R_i + R_S}$

$$R_{ms} I_s - I_0 R_0 = V_0(1 + R_{ms}\beta)$$

$$V_0 = \frac{R_{ms}}{1 + R_{ms}\beta} I_S - \frac{R_0}{1 + R_{ms}\beta} I_0$$

Output resistance with feedback is-

$$R_{of} = \frac{R_0}{1 + R_{ms}\beta}$$

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The output of the feedback element decides the type of mixing performed on the signal.

Previous Years' Question

The input impedance (Z_i) and the output impedance (Z_0) of an ideal transconductance (voltage controlled current source) amplifier are:

a) $Z_i = 0$, $Z_0 = 0$
b) $Z_i = 0$, $Z_0 = \infty$
c) $Z_i = \infty$, $Z_0 = 0$
d) $Z_i = \infty$, $Z_0 = \infty$

Due to negative feedback, both input resistance and output resistance are reduced.

For example: Collector to the base bias circuit

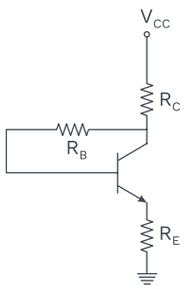


Fig. 7.9 (c) Collector to Base Bias Circuit

3) Series-series negative feedback:

This is also known as current-series feedback.

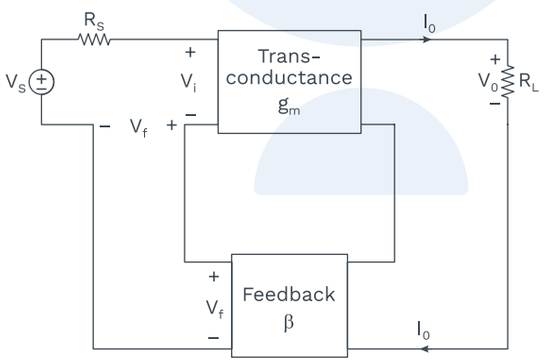


Fig. 7.10 (a) Series-Series feedback configuration

Transconductance amplifier:

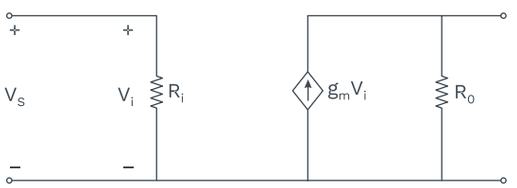


Fig. 7.10 (b) Transconductance Amplifier without feedback

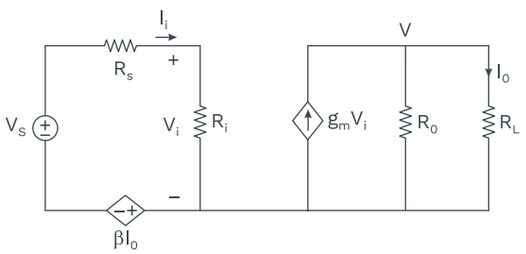


Fig. 7.10 (c) Transconductance Amplifier with feedback

Input resistance:

Applying KVL in the input loop, we get:

$$V_s - I_i R_s - I_i R_i - \beta I_0 = 0$$

$$I_0 = g_m V_i \times \frac{R_0}{R_0 + R_L} = g_{mL} V_i$$

where, g_{mL}: Transconductance considering load resistance (R_L)

$$V_s = I_i R_s + I_i R_i + \beta g_{mL} V_i$$

$$V_s = I_i R_s + I_i R_i (1 + g_{mL}\beta) \qquad \{V_i = I_i R_i\}$$

$\therefore$ Input resistance with feedback is:

$$R_{if} = (1 + g_{mL}\beta) R_i$$

Output resistance:

Applying KCL at output side, we get:

$$g_m V_i = \frac{V}{R_0} + \frac{V}{R_L}$$

$$g_m (V_s - \beta I_0) \times \frac{R_i}{R_i + R_s} = \frac{V}{R_0} + \frac{V}{R_L}$$

$$g_{ms} V_s - g_{ms} \beta I_0 = \frac{V}{R_0} + I_0$$

where, $g_{ms} = g_m \cdot \frac{R_i}{R_i + R_s}$

$$I_0 (1 + g_{ms}\beta) = g_{ms} V_s - \frac{V}{R_0}$$

$$I_0 = \frac{g_{ms} V_s}{1 + g_{ms}\beta} - \frac{V}{R_0(1 + g_{ms}\beta)}$$

$\therefore$ Output resistance with feedback is-

$$R_{0f} = R_0(1 + g_{ms}\beta)$$

For example: Common emitter amplifier with emitter resistance

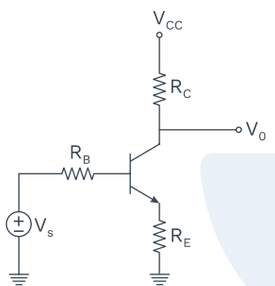


Fig. 7.10 (d) CE Amplifier with emitter resistance

4) Shunt-series negative feedback

This is also known as current-shunt feedback.

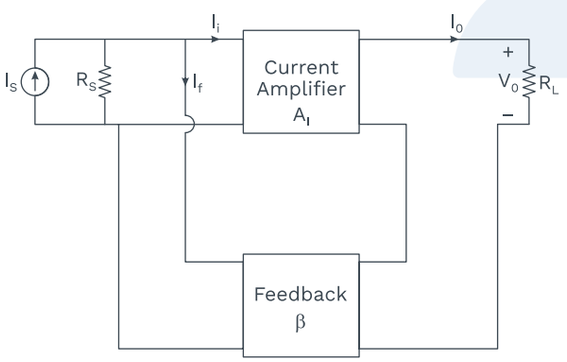


Fig. 7.11 (a) Shunt-series Feedback

The circuit of current amplifier is shown below:

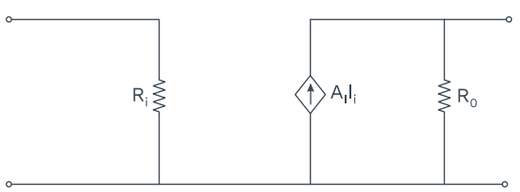


Fig. 7.11 (b) Current Amplifier without feedback

Previous Years' Question

To obtain very high input and output impedances in a feedback amplifier, the most used feedback is:

a) voltage-series
b) current-series
c) voltage-shunt
d) current-shunt

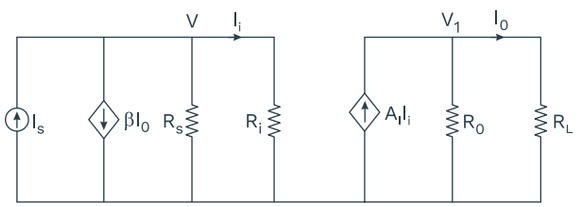


Fig. 7.11 (c) Current Amplifier with feedback

Input resistance:

Applying KCL at input side, we get:

$$I_S - \beta I_0 = \frac{V}{R_S} + \frac{V}{R_i} \qquad \text{...(i)}$$

By current division rule,

$$I_0 = (A_I I_i)\frac{R_0}{R_0 + R_L} = A_{IL} I_i \qquad \text{...(ii)}$$

where, $A_{IL} = A_I \frac{R_0}{R_0 + R_L}$

Putting the value of I_0 in the equation i), we get:

$$I_S - \beta A_{IL} I_i = \frac{V}{R_S} + \frac{V}{R_i}$$

$$I_S = \frac{V}{R_S} + \frac{V}{R_i} + \beta A_{IL} I_i$$

$$I_S = \frac{V}{R_S} + \frac{V}{R_i} + \beta A_{IL} \times \frac{V}{R_i} \quad \left\{\because I_I = \frac{V}{R_i}\right\}$$

$$I_S = \frac{V}{R_S} + \frac{V}{R_i}(1 + \beta A_{IL})$$

$\therefore$ Input resistance with feedback is:

$$R_{if} = \frac{R_i}{1 + A_{IL}\beta}$$

Output resistance:

Applying KCL at output side, we get:

$$I_0 = A_i I_i - \frac{V_1}{R_0} \qquad ...(iii)$$

By current division rule,

$$I_i = (I_s - \beta I_0)\frac{R_s}{R_i + R_s} \qquad ...(iv)$$

Putting the value of I_i in the equation iii), we get:

$$I_0 = A_I \cdot \frac{R_s}{R_i + R_s}(I_s - \beta I_0) - \frac{V_1}{R_0}$$

where, $A_{IS} = A_I \cdot \frac{R_s}{R_i + R_s}$

$$I_0 = A_{IS}(I_S - \beta I_0) - \frac{V_1}{R_0}$$

$$I_0(1 + A_{IS}\beta) = A_{IS} I_S - \frac{V_1}{R_0}$$

$$I_0 = \frac{A_{IS}}{A_{IS}\beta} I_S - \frac{V_1}{R_0(1 + A_{IS}\beta)}$$

∴ The output resistance with feedback is:

$R_{0f} = R_0(1 + A_{IS}\beta)$

Previous Years' Question

Three identical amplifiers with each one having a voltage gain of 50, the input resistance of 1 kΩ and output resistance of 250 Ω, are cascaded. The open circuit voltage gain of the combined amplifier is:

a) 49 dB **b)** 51 dB
c) 98 dB **d)** 102 dB

Source	Amplifier	Feedback	Feedback	R_{if}	R_{of}
VCVS	Voltage	Series-shunt	Voltage series	$R_i(1 + A\beta)$	$\frac{R_0}{1+A\beta}$
VCCS	Trans-conductance	Series-series	Current series	$R_i(1+A\beta)$	$R_0(1+A\beta)$
CCVS	Trans-resistance	Shunt-shunt	Voltage shunt	$\frac{R_i}{(1+A\beta)}$	$\frac{R_0}{(1+A\beta)}$
CCCS	Current	Shunt-series	Current shunt	$\frac{R_i}{(1+A\beta)}$	$R_0(1+A\beta)$

Example 3:
In comparison to the basic amplifier, the shunt-shunt negative feedback amplifier has:

a) low input and output impedances.

b) low input impedance and high output impedance.

c) high input impedance and low output impedance.

d) high input and output impedances.

Sol:
Shunt at input $\Rightarrow$ Current mixing

Shunt at output $\Rightarrow$ Voltage sampling

$\Rightarrow$ Amplifier = Trans-resistance amplifier

R_i = low; R_0 = low

Hence, the correct option is **a)**.

Example 4:
In a voltage series negative feedback amplifier, the input resistance and the output resistance, respectively are:

a) low; low

b) low; high

c) high; low

d) high; high

Sol:

Voltage – Series
↓ ↓
Sampling Mixing

Output = Voltage ⇒ Voltage mixing

Amplifier: Voltage amplifier

R_i: high, R_0 = low

Hence, the correct option is (C).

Example 5:

What kind of feedback network is depicted in the figure below?

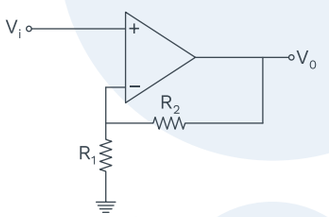


Fig. 7.12

Sol:

Equivalent circuit of the given figure:

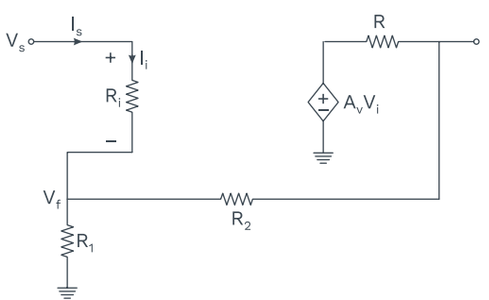


Fig. 7.13

$I_s = I_i$

$V_i = V_s - V_f$: Voltage mixing

Make $I_i = 0$, R_2 and R_1: Series

$V_f = \dfrac{V_0 R_1}{R_1 + R_2}$: Voltage sampling

So, this is shunt-series feedback network.

Example 6:
Due to series-series feedback-

a) input impedance increases, and output impedance decreases.
b) both input and output impedance increases.
c) input impedance decreases, and output impedance increases.
d) both input and output impedance decreases.

Sol:

Series – Series
↓ ↓
Mixing Sampling
(Voltage) (Current)

Transconductance amplifier
R_i = Increases R_0 = Increases
Both input and output impedance increases.
Hence, the correct option is B).

Example 7:
What is the effect of series-series feedback on input impedance and output impedance?

Sol:
Series-series ⇒ Voltage-current
Transconductance amplifier
Both input resistance (R_i) and output resistance (R_0) will increase.

Example 8:
What kind of feedback network is depicted in the figure below?

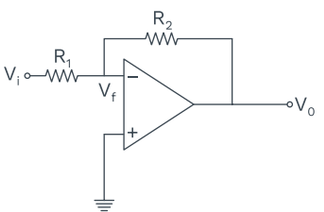


Fig. 7.14

Sol:

Equivalent circuit of the given circuit-

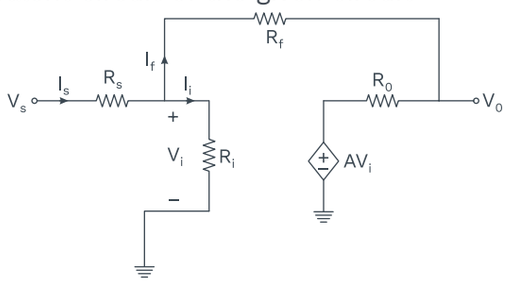


Fig. 7.15

$V_i = V_s - I_s R_s$

$I_i = I_s - I_f$ (Current mixing)

Make $V_i = 0$; $I_f = \frac{0 - V_0}{R_F} = \frac{-V_0}{R_F}$

$\Rightarrow$ Voltage sampling

So, this is shunt-shunt feedback.

Example 9:

An amplifier has an open-loop gain of 10^4 and input resistance of 50 kΩ. Find the input resistance of the series-shunt feedback amplifier with a feedback factor of 0.1.

Sol:

Input resistance, $R_{if} = R_i(1+A\beta)$

$R_i = 50\ k\Omega$

$A = 10^4$

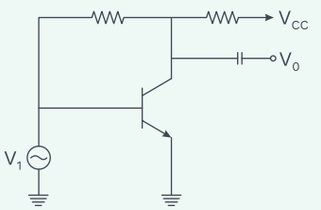
$\beta = 0.1$

$R_{if} = 50k\ (1 + 10^4 \times 0.1)$

$R_{if} \approx 50\ M\Omega$

Previous Years' Question

The circuit of the figure is an example of feedback of which of the following type?



a) current series
b) current shunt
c) voltage series
d) voltage shunt

Chapter Summary



- A feedback amplifier is one in which the output is fed back to the input.
- When an output signal is applied to the input with phase change, it is known as negative feedback.
- When an output signal is applied to the input without any phase change, it is known as positive feedback.
- There are various advantages of using negative feedback:
  i) Stabilizes the voltage gain
  ii) Increases the bandwidth
- The circuit which is used to sample the output and connect it back to the feedback network is known as the sampling circuit.
- There are two types of sampling circuit:
  i) Shunt sampling or voltage sampling
  ii) Current sampling or series sampling
- In a shunt sampling circuit, the feedback factor is unitless.
- A mixer network is a network that is used to mix the input signals with the feedback signal.
  There are two types of mixer networks:
  i) Series mixing
  ii) Shunt mixing
- Positive feedback is also known as regenerative feedback.
- Negative feedback is also known as degenerative feedback.
- The sensitivity of a device is nothing but represents how much a device is affected due to the external changes.
- There are basically four types of feedback topology:
  i) Series-shunt feedback.
  ii) Shunt-series feedback.
  iii) Series-series feedback.
  iv) Shunt-shunt feedback.
- Emitter follower employs voltage series feedback.

PRACTICE QUESTIONS

Q1 **An amplifier has an open gain of 100 and its lower and upper cut-off frequencies are 100 Hz and 100 kHz, respectively. If a feedback network with a feedback factor of 0.99 is connected to an amplifier, the lower and upper cut-off frequency respectively are:**

a) 10 Hz and 10 MHz **b) 1 Hz and 10 MHz**
c) 1 Hz and 1 MHz **d) 10 Hz and 1 MHz**

Q2 **The circuit is shown in the figure below.**

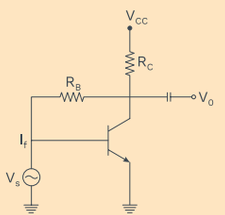


Fig. 7.17 Feedback Amplifier Circuit

The feedback used in the circuit is-

a) Shunt-shunt **b) Shunt-series**
c) Series-shunt **d) Series-series**

Q3 **The amplifier in the voltage series mode has the following parameters:**
$A = 100$, $R_i = 1\ k\Omega$, $R_0 = 100\Omega$, $\beta = 0.99$
Find the input impedance of feedback amplifier.

a) 10 kΩ **b) 20 kΩ** **c) 100 kΩ** **d) 10 MΩ**

Q4 **The negative feedback amplifier has A = –1000 and $\beta = -\frac{1}{20}$. Calculate the gain of the amplifier.**

a) 1020 **b) 50** **c) 1000** **d) $\frac{1000}{51}$**

Q5 **The feedback in the amplifier circuit shown in the figure is-**

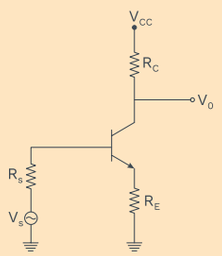


Fig. 7.18 Feedback Amplifier Circuit

a) Shunt-shunt **b) Series-shunt**
c) Shunt-series **d) Series-series**

Q6 **An AC equivalent circuit is shown in the figure. If I_i is the input current and R_f is very large, then the nature of feedback is-**

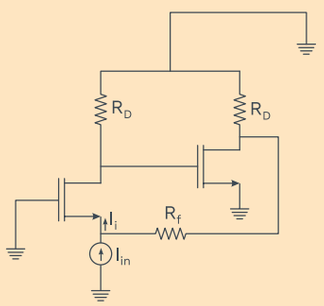


Fig. 7.19 Feedback Amplifier Circuit

a) Series-series **b) Shunt-series**
c) Shunt-shunt **d) Series-shunt**

Q7 **The loop gain of a feedback amplifier is-**

a) $-A\beta$ **b) $A\beta$** **c) $1 + A\beta$** **d) $1 - A\beta$**

Q8 **The upper cut-off frequency of a feedback amplifier is a factor of:**
a) Aβ **b) 1 + Aβ** **c) 1 – Aβ** **d) A/β**

Q9 **Which of the following type of topology is there in the cascading amplifier?**
a) Current shunt **b) Voltage shunt**
c) Current series **d) Voltage series**

Q10 **Which of the following is not the effect of negative feedback?**
a) Increase bandwidth **b) Increase input impedance**
c) Increase output impedance **d) Decrease harmonic distortion**

Answer Key:

1) b)	**2) a)**	**3) c)**	**4) d)**
5) d)	**6) c)**	**7) a)**	**8) b)**
9) a)	**10) c)**		